Testing the Substitution Predictions of Demand Models: Evidence from the Blue Bell Recall

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Abstract

Demand models are widely used in empirical industrial organization, yet evidence on their ability to predict real-world counterfactuals remains limited. I use the 2015 Blue Bell ice cream recall, which removed a major regional brand from grocery shelves for several months, to test how model structure and identifying information shape substitution predictions. I compare the predictions of nested logit, random-coefficients logit, and random-coefficients nested logit models with observed changes in rival sales. Random coefficients do not make nest structure less important: the nest structure and the strength of within-nest correlation both change predictions substantially, and models combining nests with random coefficients give some of the most accurate predictions. Holding the model fixed, household category-purchase moments improve predictions in markets where Blue Bell was large, with or without nesting instruments, and using both gives the lowest prediction loss. The results favor comparing economically motivated nest structures even in random-coefficients models and estimating them with household purchase information.

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